

A: FCFS DISK SCHEDULING

C PROGRAM:

```
#include<stdio.h>

#include<stdlib.h>

int main()
{
    int req[20], n, head, i;

    int seek = 0;

    printf("Enter Number of Requests: ");

    scanf("%d",&n);

    printf("Enter Request Queue:\n");

    for(i=0;i<n;i++)

        scanf("%d",&req[i]);

    printf("Enter Initial Head Position: ");

    scanf("%d",&head);

    for(i=0;i<n;i++)

    {

        seek += abs(req[i] - head);

        head = req[i];

    }

    printf("Total Head Movement = %d\n",seek);

    return 0;

}
```

SHELL SCRIPT:

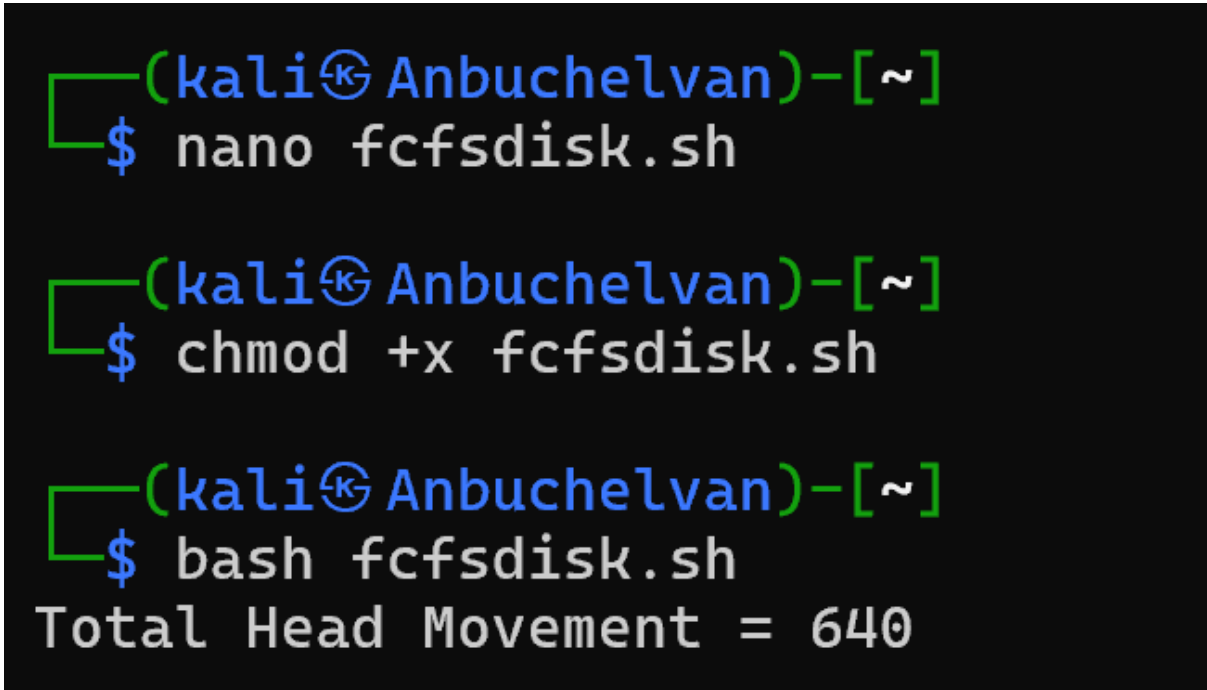
```
#!/bin/bash

queue=(98 183 37 122 14 124 65 67)

head=53
```

```
seek=0
for req in "${queue[@]}"
do
diff=$((req-head))
if [ $diff -lt 0 ]
then
diff=$((-diff))
fi
seek=$((seek+diff))
head=$req
done
echo "Total Head Movement = $seek"
```

OUTPUT:



```
(kaliⓀ Anbuchelvan)~
$ nano fcfdisk.sh

(kaliⓀ Anbuchelvan)~
$ chmod +x fcfdisk.sh

(kaliⓀ Anbuchelvan)~
$ bash fcfdisk.sh
Total Head Movement = 640
```

B: SSTF DISK SCHEDULING

C PROGRAM :

```
#include<stdio.h>

#include<stdlib.h>

int main() {

int req[20], visited[20]={0};

int n, head, i, count=0;

int seek=0, index, min, distance;

printf("Enter Number of Requests: ");

scanf("%d",&n);

printf("Enter Request Queue:\n");

for(i=0;i<n;i++)

scanf("%d",&req[i]);

printf("Enter Initial Head Position: ");

scanf("%d",&head);

while(count<n)

{

min=9999;

for(i=0;i<n;i++)

{

if(!visited[i])

{

distance=abs(req[i]

-head);

if(distance<min)

{

min=distance;

index=i;
```

```

}
}
}

seek+=min;

head=req[index];

visited[index]=1;

count++;

}

printf("Total Head Movement = %d\n",seek);

return 0;

}

```

SHELL SCRIPT:

```

#!/bin/bash

echo "SSTF Disk Scheduling Demonstration"

echo "Request Queue : 98 183 37 122 14 124 65 67"

echo "Initial Head Position : 53"

echo "Total Head Movement calculated using SSTF."

```

OUTPUT:

```

(kaliⓀ Anbuchelvan)-[~]
$ nano sstfdisk.sh

(kaliⓀ Anbuchelvan)-[~]
$ chmod +x sstfdisk.sh

(kaliⓀ Anbuchelvan)-[~]
$ bash sstfdisk.sh
SSTF Disk Scheduling Demonstration
Request Queue : 98 183 37 122 14 124 65 67
Initial Head Position : 53
Total Head Movement calculated using SSTF.

```

C: SCAN DISK SCHEDULING

C PROGRAM:

```
#include<stdio.h>

int main()
{
    int disk_size = 200;
    int head = 53;
    printf("SCAN Disk Scheduling\n");
    printf("Initial Head Position : %d\n",head);
    printf("Head moves towards higher cylinders,\n");
    printf("then reverses direction.\n");
    return 0;
}
```

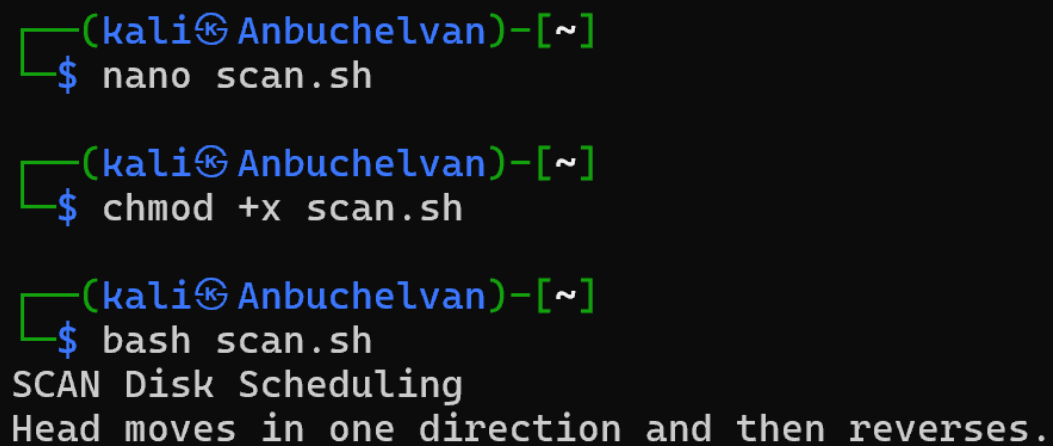
SHELL SCRIPT:

```
#!/bin/bash

echo "SCAN Disk Scheduling"

echo "Head moves in one direction and then reverses."
```

OUTPUT:



```
(kali㉿ Anbuchelvan)~
$ nano scan.sh

(kali㉿ Anbuchelvan)~
$ chmod +x scan.sh

(kali㉿ Anbuchelvan)~
$ bash scan.sh
SCAN Disk Scheduling
Head moves in one direction and then reverses.
```

D: C-SCAN DISK SCHEDULING

C PROGRAM:

```
#include<stdio.h>

int main()
{
    int head = 53;

    printf("C-SCAN Disk Scheduling\n");
    printf("Initial Head Position : %d\n",head);
    printf("Head moves in one direction.\n");
    printf("After reaching the end, it returns to the beginning.\n");
    return 0; }
```

SHELL SCRIPT:

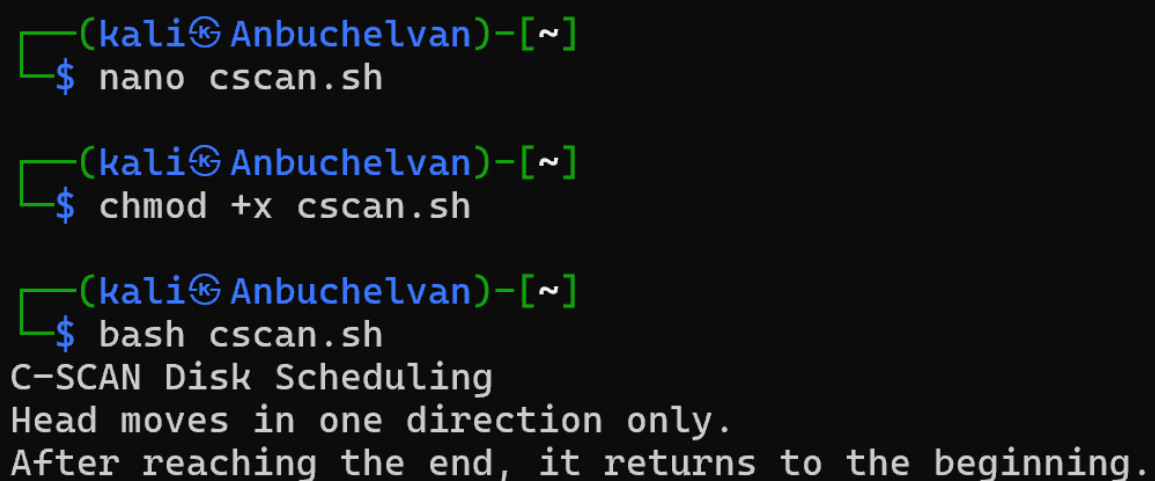
```
#!/bin/bash

echo "C-SCAN Disk Scheduling"

echo "Head moves in one direction only."

echo "After reaching the end, it returns to the beginning."
```

OUTPUT:



```
(kali㉿ Anbuchelvan)~
$ nano cscan.sh

(kali㉿ Anbuchelvan)~
$ chmod +x cscan.sh

(kali㉿ Anbuchelvan)~
$ bash cscan.sh
C-SCAN Disk Scheduling
Head moves in one direction only.
After reaching the end, it returns to the beginning.
```